

# Практикум C7:archive

Twenty questions, 99 marks. Every question the archive measures with vectors, from May 2021 to November 2025, in the order of the seven techniques rather than the order of the papers.

## О практикуме

Третий и последний практикум по векторам, и остаток после C5 и C6 оказался однородным: здесь векторами меряют. Площадь, объём, угол, расстояние — четыре величины, и все четыре берутся из векторного произведения. 48 баллов из 182 geometry.vectors (0.264) и 66 из 216 geometry.vectors\_3d (0.3053). 114 баллов по корпусу, 108 рабочих — ноябрь 2023 Q8(a), (c) стоит дважды. 37 баллов Paper 1, 56 Paper 2, 21 Paper 3: единственный практикум по векторам, где Paper 3 весит.

## Заданий

0

## Раздел решений

включён

## **Интерактивная версия**

[kaggle.com/code/artemkonukhov/ib-math-praktikum-c7-measuring](https://kaggle.com/code/artemkonukhov/ib-math-praktikum-c7-measuring)

Собрано из ноутбука репозитория [ib-math-aa-hl-past-papers](#). Проверки в PDF не работают — они для интерактивной версии.

# C7 archive — measuring in space, all of it

**Twenty questions, 99 marks.** Every question the archive measures with vectors, from May 2021 to November 2025, in the order of the seven techniques rather than the order of the papers.

No theory. No worked examples. The theory is in the practicum, `practicum-c7-measuring.ipynb`.

| § | technique                   | questions | marks |
|---|-----------------------------|-----------|-------|
| 1 | The vector product          | 2         | 6     |
| 2 | Area                        | 4         | 19    |
| 3 | Volume                      | 1         | 4     |
| 4 | The identity                | 3         | 19    |
| 5 | Angles with a plane         | 6         | 30    |
| 6 | The closest point on a line | 2         | 12    |
| 7 | The distance to a line      | 2         | 9     |

The checks are the same ones the practicum uses and they store nothing: each is handed the figure exactly as the

question names it — `verify_measure('...', q, 'triangle', A, B, C)`, `verify_cross('...', q, B - A, C - A)`, `verify_distance('...', q, l1, l2)`, `verify_angle('...', q, P1, P2)` — and a point the question describes in words is handed **the conditions** that fix it: `on(N, L)`, `perpendicular(N - point_P, L)`. Three significant figures, unless the question asks otherwise.

**Parts of one question are split by technique.** November 2023 Q8 appears in §§ 1 and 2; May 2024 TZ1 Q12 in §§ 2 and 4; the Paper 3 sphere of May 2025 TZ2 in §§ 1 and 5. Each part repeats what it needs.

**A *show that* is checked by the line before the given result:** the vector product itself before it is compared with the printed column, the angle before it is called right, the left-hand side of the identity written with lengths and the angle.

Enter a vector as `vec(a, b, c)` or `(a, b, c)`; `lam` and `mu` are  $\lambda$  and  $\mu$ .

Solutions are at the very bottom, deliberately far away.

```
import sys
sys.path.append('.')          # from
practicum/geometry to practicum/kit/
import sympy as sp           # the escape
hatch: anything not in kit is in sp
from kit import *            # checks +
vec, cross, measure, plane, unknown

language('en')               # this
notebook is in English, and so are the checks

print('ready; sympy', sp.__version__)
```

---

# § 1. The vector product

**Two questions, 6 marks.** Two vectors in, one vector out — with a letter in the components and without.

## 1.1 — November 2023 TZ1 Paper 2 Q8(a), 4 marks

Three points are given by  $A(0, p, 2)$ ,  $B(1, 1, 1)$  and  $C(p, 0, 4)$ , where  $p$  is a positive constant.

Show that  $\overrightarrow{AB} \times \overrightarrow{AC} = \begin{pmatrix} 2 - 3p \\ -2 - p \\ p^2 - 2p \end{pmatrix}$ .

*Enter the product you computed, before comparing it with the printed column.*

```
p = symbols('p', positive=True)
A, B, C = vec(0, p, 2), vec(1, 1, 1), vec(p,
0, 4)

q1_1 = ...          # AB × AC

verify_cross('1.1', q1_1, B - A, C - A)
```

## 1.2 — May 2025 TZ2 Paper 3 Q2(c)(i), 2 marks

A sphere of radius 6 is centred at the origin  $O$ ; lengths are in thousands of kilometres. The North Pole  $P$  and a point  $A$  on the equator have position vectors

$$\mathbf{p} = \begin{pmatrix} 0 \\ 0 \\ 6 \end{pmatrix} \text{ and } \mathbf{a} = \begin{pmatrix} 6 \\ 0 \\ 0 \end{pmatrix}.$$

Find the vector  $\mathbf{a} \times \mathbf{p}$ .

```
a, p_pole = vec(6, 0, 0), vec(0, 0, 6)
```

```
q1_2 = ...          # a × p
```

```
verify_cross('1.2', q1_2, a, p_pole)
```

---

## § 2. Area

**Four questions, 19 marks.** The length of a vector product is the area of the parallelogram on its two factors; a triangle is half of it. With a letter in the points the area becomes a function, and *smallest* means its minimum.

### **2.1 — November 2023 TZ1 Paper 2 Q8(b), (c), 5 marks**

Continuing 1.1, with  $A(0, p, 2)$ ,  $B(1, 1, 1)$ ,  $C(p, 0, 4)$  and  $p > 0$ :

**(a)** Hence, find the smallest possible value of  $|\overrightarrow{AB} \times \overrightarrow{AC}|^2$ .

**(b)** Hence, find the smallest possible area of triangle  $ABC$ .

*Three significant figures.*

```

p = symbols('p', positive=True)
A, B, C = vec(0, p, 2), vec(1, 1, 1), vec(p,
0, 4)
product = cross(B - A, C - A)

q2_1a = ...      # the smallest  $|AB \times AC|^2$ 
q2_1b = ...      # the smallest area of ABC

verify_optimum('2.1(a)', q2_1a, dot(product,
product), p, Interval(0, oo), 'min')
verify_optimum('2.1(b)', q2_1b,
measure('triangle', A, B, C), p, Interval(0,
oo), 'min')

```

## 2.2 — May 2025 TZ3 Paper 1 Q11(c), 4 marks

The points  $A(1, -4, 0)$ ,  $B(-3, -6, 2)$ ,  $C(-1, -2, 4)$  and  $D(3, 0, 2)$  form a parallelogram  $ABCD$ , where  $D$  is diagonally opposite  $B$ .

**(a)** Given that  $\overrightarrow{AB} \times \overrightarrow{AD} = m \begin{pmatrix} -1 \\ 1 \\ -1 \end{pmatrix}$ ,

where  $m \in \mathbb{Z}^+$ , find the value of  $m$ .

**(b)** Hence, find the area of parallelogram  $ABCD$ .

*Give (b) exactly.*

```

m = symbols('m')
A, B, C, D = vec(1, -4, 0), vec(-3, -6, 2),
vec(-1, -2, 4), vec(3, 0, 2)

q2_2a = ...      # m
q2_2b = ...      # the area of ABCD

verify_find('2.2(a)', q2_2a, m, [Eq(cross(B -
A, D - A), m*vec(-1, 1, -1))])
verify_measure('2.2(b)', q2_2b,
'parallelogram', A, B, C, D, exact=True)

```

### **2.3 — November 2025 TZ1 Paper 2 Q12(d), 5 marks**

The plane  $\Pi$  has Cartesian equation  $4x + y - 3z = 18$  and meets the coordinate axes at  $P(4.5, 0, 0)$ ,  $Q(0, 18, 0)$  and  $R(0, 0, -6)$ .

Use a vector method to find the area of the triangle  $PQR$ .

*Three significant figures.*

```

point_P, Q, R = vec(Rational(9, 2), 0, 0),
vec(0, 18, 0), vec(0, 0, -6)

q2_3 = ...      # the area of PQR

verify_measure('2.3', q2_3, 'triangle',
point_P, Q, R)

```

## 2.4 — May 2024 TZ1 Paper 2 Q12(c), 5 marks

A triangle  $ABC$  has vertices  $A(0, 1, 2)$ ,  $B(1, 2, 3)$  and  $C(3, 2, 1)$ , and  $\mathbf{u} = \overrightarrow{AB}$ ,  $\mathbf{v} = \overrightarrow{AC}$ . Consider a new point  $D$ , and let  $\mathbf{w} = \overrightarrow{CD}$ .

It is given that  $\mathbf{u} \cdot \mathbf{w} = \mathbf{v} \cdot \mathbf{w} = 0$  and that the area of triangle  $ACD$  is 5 square units.

Find the possible vectors for  $\mathbf{w}$ .

*Enter both, as a list of two vectors.*

```
A, B, C = vec(0, 1, 2), vec(1, 2, 3), vec(3, 2, 1)
u, v = B - A, C - A
w = unknown('w', 3)
D = C + w

q2_4 = [...]      # both possible w

verify_find('2.4', q2_4, w, [perpendicular(w, u), perpendicular(w, v),
Eq(measure('triangle', A, C, D), 5)])
```

## § 3. Volume

**One question, 4 marks.** A pyramid is a third of base times height — or a sixth of the determinant of its three edges from the apex.

### 3.1 — November 2025 TZ1 Paper 2 Q12(g), 4 marks

Continuing 2.3: the plane  $\Pi$  meets the axes at  $P(4.5, 0, 0)$ ,  $Q(0, 18, 0)$  and  $R(0, 0, -6)$ , and the line  $L_3$ , normal to  $\Pi$  through  $(1, 8, -2)$ , contains the point  $S(-11, 5, 7)$  at  $\gamma = 3$ .

Hence, find the volume of pyramid  $PQRS$ .

*The answer is a whole number.*

```
point_P, Q, R = vec(Rational(9, 2), 0, 0),  
vec(0, 18, 0), vec(0, 0, -6)  
S = vec(-11, 5, 7)
```

```
q3_1 = ...          # the volume of PQRS
```

```
verify_measure('3.1', q3_1, 'pyramid', S,  
point_P, Q, R, exact=True)
```

## § 4. The identity

**Three questions, 19 marks.**  $(\mathbf{u} \cdot \mathbf{v})^2 + |\mathbf{u} \times \mathbf{v}|^2 = |\mathbf{u}|^2 |\mathbf{v}|^2$  — and the length it hands you when nobody gives it.

### **4.1 — May 2021 TZ2 Paper 1 Q5, 4 marks**

Given any two non-zero vectors  $\mathbf{a}$  and  $\mathbf{b}$ , show that  $|\mathbf{a} \times \mathbf{b}|^2 = |\mathbf{a}|^2 |\mathbf{b}|^2 - (\mathbf{a} \cdot \mathbf{b})^2$ .

*Enter the two ends of the proof: the left-hand side and the last term of the right-hand side, each written with  $A = |\mathbf{a}|$ ,  $B = |\mathbf{b}|$  and  $\theta$ . Use `th` for  $\theta$ .*

```

A, B, th = symbols('A B th')
a = vec(*symbols('a1 a2 a3'))          #
any two vectors, in components
b = vec(*symbols('b1 b2 b3'))
letters = {A: mag(a), B: mag(b), th: angle(a,
b)}

q4_1a = ...          #  $|a \times b|^2$ , written with A,
B and th
q4_1b = ...          #  $(a \cdot b)^2$ , written with A,
B and th

verify_formula('4.1(a)', q4_1a, dot(cross(a,
b), cross(a, b)), letters)
verify_formula('4.1(b)', q4_1b, dot(a, b)**2,
letters)

```

## 4.2 — May 2024 TZ1 Paper 2 Q12(a), 2 marks

Consider the non-zero vectors  $\mathbf{u}$  and  $\mathbf{v}$ , and let  $\theta$  be the angle between them. Using the definitions of  $\mathbf{u} \cdot \mathbf{v}$  and  $\mathbf{u} \times \mathbf{v}$  in terms of  $|\mathbf{u}|$ ,  $|\mathbf{v}|$  and  $\theta$ , show that  $(\mathbf{u} \cdot \mathbf{v})^2 + |\mathbf{u} \times \mathbf{v}|^2 = |\mathbf{u}|^2 |\mathbf{v}|^2$ .

*Enter the left-hand side written with  $U = |\mathbf{u}|$ ,  $V = |\mathbf{v}|$  and  $th$  — before you simplify it.*

```

U, V, th = symbols('U V th')
one = vec(*symbols('m1 m2 m3'))          #
any two vectors, in components
two = vec(*symbols('n1 n2 n3'))

q4_2 = ...      #  $(\mathbf{u} \cdot \mathbf{v})^2 + |\mathbf{u} \times \mathbf{v}|^2$ ,
written with U, V and th

verify_formula('4.2', q4_2, dot(one, two)**2
+ dot(cross(one, two), cross(one, two)),
      {U: mag(one), V: mag(two), th:
angle(one, two)})

```

### 4.3 — May 2024 TZ1 Paper 2 Q12(b), 13 marks

A triangle  $ABC$  has vertices  $A(0, 1, 2)$ ,  $B(p, q, 3)$  and  $C(3, 2, 1)$ , with  $p, q \in \mathbb{R}$ . The vectors are  $\mathbf{u} = \overrightarrow{AB}$  and  $\mathbf{v} = \overrightarrow{AC}$ . It is given that  $\mathbf{u} \cdot \mathbf{v} = 3$  and that the area of triangle  $ABC$  is  $\sqrt{6}$ .

- (a)** Find the value of  $|\mathbf{u} \times \mathbf{v}|$ .
- (b)** Hence, or otherwise, find the value of  $|\mathbf{u}|$ .
- (c)** Hence, or otherwise, find the possible values of  $p$  and the corresponding values of  $q$ .

*For (c) enter both pairs, as `[[p, q], [p, q]]`.*

```
p, q = unknown('p'), unknown('q')           #
the two the question asks for
A, B, C = vec(0, 1, 2), vec(p, q, 3), vec(3,
2, 1)
u, v = B - A, C - A
given = [Eq(measure('triangle', A, B, C),
sqrt(6)), Eq(dot(u, v), 3)]

q4_3a = ...           # |u × v|
q4_3b = ...           # |u|
q4_3c = [...]         # [[p, q], [p, q]]

verify_find('4.3(a)', q4_3a, mag(cross(u,
v)), given)
verify_find('4.3(b)', q4_3b, mag(u), given)
verify_find('4.3(c)', q4_3c, [p, q], given)
```

## § 5. Angles with a plane

**Six questions, 30 marks.** A plane has no direction, only a normal: the angle between two planes is the acute angle between their normals, and the angle a line makes with a plane is what is left of a right angle. On a sphere the same rule appears between two vector products.

### **5.1 — May 2025 TZ1 Paper 1 Q11(a), 6 marks**

The plane  $P_1$  has equation  $x + 2y + z = 0$  and the plane  $P_2$  has equation  $x - y - 2z = 0$ . The acute angle between the planes  $P_1$  and  $P_2$  is  $\theta$ .

Show that  $\theta = 60^\circ$ .

*Enter the angle in degrees.*

```
P1 = plane(Eq(x + 2*y + z, 0), name='P1')
P2 = plane(Eq(x - y - 2*z, 0), name='P2')

q5_1 = ...          #  $\theta$  in degrees

verify_angle('5.1', q5_1, P1, P2, deg=True)
```

## 5.2 — May 2025 TZ3 Paper 1 Q11(e), 3 marks

The plane  $P_1$  contains the parallelogram  $ABCD$  of 2.2 and has equation  $-x + y - z = -5$ . A second plane  $P_2$  has equation  $5x + y - 7z = 1$ . The acute angle between  $P_1$  and  $P_2$  is  $\theta$ .

Show that  $\cos \theta = \frac{1}{5}$ .

*Exactly.*

```
P1 = plane(Eq(-x + y - z, -5), name='P1')
P2 = plane(Eq(5*x + y - 7*z, 1), name='P2')

q5_2 = ...          # cos θ

verify_angle('5.2', q5_2, P1, P2,
cosine=True, exact=True)
```

## 5.3 — May 2023 TZ1 Paper 2 Q8, 7 marks

The angle between a line and a plane is  $\alpha$ , where  $\alpha \in \mathbb{R}$ ,  $0 < \alpha < \frac{\pi}{2}$ . The equation of the line is  $\frac{x-1}{3} = \frac{y+2}{2} = 5-z$ , and the equation of the plane is  $4x + (\cos \alpha)y + (\sin \alpha)z = 1$ .

Find the value of  $\alpha$ .

*Three significant figures, in radians.*

```
alpha = symbols('alpha', positive=True)
L = cartesian((x - 1)/3, (y + 2)/2, 5 - z)
Pi = plane(Eq(4*x + cos(alpha)*y +
sin(alpha)*z, 1))

q5_3 = ...          #  $\alpha$  in radians

verify_find('5.3', q5_3, alpha, [Eq(angle(L,
Pi), alpha)])
```

### **5.4 — May 2025 TZ2 Paper 3 Q2(c) (ii), 3 marks**

On the sphere of 1.2, with  $\mathbf{p} = (0, 0, 6)$ ,  $\mathbf{n} = (0, 6, 0)$  and  $\mathbf{a} = (6, 0, 0)$ :  $P$ ,  $N$  and  $A$ , and the arcs joining them, form a spherical triangle, and the angle at the vertex  $A$  is defined as the angle between  $\mathbf{a} \times \mathbf{p}$  and  $\mathbf{a} \times \mathbf{n}$ .

Show that the angle at vertex  $A$  in the spherical triangle is  $90^\circ$ .

*Enter the angle in degrees.*

```
a, p_pole, n = vec(6, 0, 0), vec(0, 0, 6),  
vec(0, 6, 0)
```

```
q5_4 = ...          # the angle at A, in degrees
```

```
verify_angle('5.4', q5_4, cross(a, p_pole),  
cross(a, n), deg=True)
```

### **5.5 — May 2025 TZ2 Paper 3 Q2(e), 5 marks**

On the same sphere, Moscow  $M$  has  $\mathbf{m} = \begin{pmatrix} 0 \\ 6 \cos \theta \\ 6 \sin \theta \end{pmatrix}$  with  $\theta = 57.3^\circ$ , and Bogotá  $B$  has  $\mathbf{b} = \begin{pmatrix} 6 \sin 120^\circ \\ 6 \cos 120^\circ \\ 0 \end{pmatrix}$ .

Find the shortest distance from Bogotá to Moscow on the sphere.

*In thousands of kilometres, three significant figures.*

```

theta = 57.3*pi/180                                #
57.3 degrees, from part (d)
b = vec(6*sin(2*pi/3), 6*cos(2*pi/3), 0)          #
120 degrees west of Nairobi
m = vec(0, 6*cos(theta), 6*sin(theta))

q5_5 = ...          # the distance along the
sphere

verify_arc('5.5', q5_5, 6, b, m)

```

## 5.6 — May 2025 TZ2 Paper 3 Q2(f), 6 marks

The bearing from  $B$  to  $M$  is the angle at the vertex  $B$  in the spherical triangle  $BMP$  — that is, by the method of 5.4, the angle between  $\mathbf{b} \times \mathbf{m}$  and  $\mathbf{b} \times \mathbf{p}$ .

Using the method from part (c), find the bearing from Bogotá to Moscow.

*In degrees, three significant figures.*

```

theta = 57.3*pi/180
b = vec(6*sin(2*pi/3), 6*cos(2*pi/3), 0)
m = vec(0, 6*cos(theta), 6*sin(theta))
p_pole = vec(0, 0, 6)

q5_6 = ...          # the bearing, in degrees

verify_angle('5.6', q5_6, cross(b, m),
cross(b, p_pole), deg=True)

```



## § 6. The closest point on a line

**Two questions, 12 marks.** The point of a line nearest a given point is the one where the segment to it crosses the line at a right angle.

### 6.1 — November 2025 TZ3 Paper 1 Q10(d), (e), 7 marks

The point  $P(-1, 1, -13)$  lies on the line  $L_1$ . The line  $L_2$  passes through  $A(2, -4, 2)$  and  $B(7, -6, 1)$ , so  $\mathbf{s} = \begin{pmatrix} 2 \\ -4 \\ 2 \end{pmatrix} + \mu \begin{pmatrix} 5 \\ -2 \\ -1 \end{pmatrix}$ . The point  $N$  lies on  $L_2$ .

**(a)** Find  $\overrightarrow{PN} \cdot \overrightarrow{AB}$  in terms of  $\mu$ .

**(b)** Given that  $N$  is the point on  $L_2$  that lies closest to  $P$ , find the coordinates of  $N$ .

Use `mu` for  $\mu$ ; give (b) exactly.

```

point_P, A, B = vec(-1, 1, -13), vec(2, -4,
2), vec(7, -6, 1)
L2 = line(A + mu*(B - A), mu, name='L2')
N = unknown('N', 3)

q6_1a = ...      # PN · AB, in terms of mu
q6_1b = ...      # N

verify_find('6.1(a)', q6_1a, dot(L2.at(mu) -
point_P, B - A))
verify_find('6.1(b)', q6_1b, N, [on(N, L2),
perpendicular(N - point_P, L2)])

```

## 6.2 — May 2023 TZ2 Paper 2 Q6(b), 5 marks

$L$  is the line of intersection of the planes  $\Pi_1 : 2x - y + 2z = 6$  and  $\Pi_2 : 4x + 3y - z = 2$ ,

and a vector equation of it is  $\mathbf{r} = \begin{pmatrix} 0 \\ 2 \\ 4 \end{pmatrix} + \lambda \begin{pmatrix} 1 \\ -2 \\ -2 \end{pmatrix}$ .

Find the coordinates of the point  $P$  on  $L$  that is nearest to the origin.

*Exactly.*

```
L = line(vec(0, 2, 4) + lam*vec(1, -2, -2),  
lam, name='L')  
closest = unknown('N', 3)  
  
q6_2 = ...      # the point of L nearest to  
the origin  
  
verify_find('6.2', q6_2, closest,  
[on(closest, L), perpendicular(closest, L)])
```

---

## § 7. The distance to a line

**Two questions, 9 marks.** The shortest distance to a line is the length of the perpendicular —  $|\overrightarrow{AP} \times \mathbf{v}|/|\mathbf{v}|$ , and dividing by  $|\mathbf{v}|$  is not optional.

### **7.1 — May 2023 TZ1 Paper 1 Q12(c), 4 marks**

Two lines  $L_1$  and  $L_2$  meet at  $P$ , and  $A(2t, 8, 3)$  lies on  $L_2$ , where  $t > 0$ . The acute angle between the lines is  $\frac{\pi}{3}$ , the direction vector of  $L_1$  is  $(1, 1, 0)$ , and  $\overrightarrow{PA} = (2t, 0, 3 + t)$ . Parts (a) and (b) give  $t = 3$ , so  $P(0, 8, -3)$  and  $A(6, 8, 3)$ .

Hence or otherwise, find the shortest distance from  $A$  to  $L_1$ .

*Exactly.*

```

L1 = line(vec(0, 8, -3) + lam*vec(1, 1, 0),
lam, name='L1')
A = vec(6, 8, 3)

q7_1 = ...          # the shortest distance from
A to L1

verify_distance('7.1', q7_1, A, L1,
exact=True)

```

## 7.2 — May 2021 TZ2 Paper 1 Q8(b), 5 marks

The lines  $l_1$  and  $l_2$  have vector equations

$$\begin{aligned}
 l_1 : \mathbf{r}_1 &= \begin{pmatrix} 3 \\ 2 \\ -1 \end{pmatrix} + \lambda \begin{pmatrix} 2 \\ -2 \\ 2 \end{pmatrix} & l_2 : \mathbf{r}_2 \\
 &= \begin{pmatrix} 2 \\ 0 \\ 4 \end{pmatrix} + \mu \begin{pmatrix} 1 \\ -1 \\ 1 \end{pmatrix}
 \end{aligned}$$

Part (a) shows that  $l_1$  and  $l_2$  do not intersect.

Find the minimum distance between  $l_1$  and  $l_2$ .

*Exactly.*

```
l1 = line(vec(3, 2, -1) + lam*vec(2, -2, 2),  
lam, name='l1')  
l2 = line(vec(2, 0, 4) + mu*vec(1, -1, 1),  
mu, name='l2')  
  
q7_2 = ...          # the minimum distance  
between l1 and l2  
  
verify_distance('7.2', q7_2, l1, l2,  
exact=True)
```

---

---



# Solutions

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## § 1

$$\mathbf{1.1} \quad \overrightarrow{AB} = (1, 1 - p, -1), \quad \overrightarrow{AC} = (p, -p, 2):$$

$$\boxed{(2 - 3p, -2 - p, p^2 - 2p)}.$$

$$\mathbf{1.2} \quad (6, 0, 0) \times (0, 0, 6) = \boxed{(0, -36, 0)}.$$


---

## § 2

$$\mathbf{2.1} \quad (\mathbf{a}) \quad (2 - 3p)^2 + (2 + p)^2 + (p^2 - 2p)^2 =$$

$$p^4 - 4p^3 + 14p^2 - 8p + 8, \quad \text{least at } p =$$

$$0.3264\dots: \boxed{6.75}. \quad (\mathbf{b}) \quad \frac{1}{2}\sqrt{6.75257\dots} = \boxed{1.30}$$

units<sup>2</sup>.

$$\mathbf{2.2} \quad (\mathbf{a}) \quad (-4, -2, 2) \times (2, 4, 2) = (-12, 12, -12):$$

$$\boxed{m = 12}. \quad (\mathbf{b}) \quad 12\sqrt{3} = \boxed{12\sqrt{3}} \approx 20.8.$$

$$\mathbf{2.3} \quad \overrightarrow{PQ} \times \overrightarrow{PR} = (-108, -27, 81), \quad \text{so}$$

$$\frac{1}{2}\sqrt{18954} = \boxed{68.8}.$$

$$\mathbf{2.4} \quad \mathbf{w} \parallel \mathbf{u} \times \mathbf{v} = (-2, 4, -2), \quad \text{and } \frac{1}{2}|\mathbf{v}||\mathbf{w}| = 5$$

gives  $|\mathbf{w}| = \frac{10}{\sqrt{11}}:$

$$\boxed{\pm \frac{5\sqrt{66}}{33}(1, -2, 1)} \approx$$

$$\pm(1.23, -2.46, 1.23).$$

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## § 3

**3.1** Height  $3|(-4, -1, 3)| = \sqrt{234}$ , base 68.8:  
 $\frac{1}{3} \cdot \frac{1}{2} \sqrt{18954} \cdot \sqrt{234} = \boxed{351}$ .

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## § 4

**4.1 (a)**  $\boxed{A^2 B^2 \sin^2 \theta}$ . **(b)**  $\boxed{A^2 B^2 \cos^2 \theta}$ , and  
 $A^2 B^2 \sin^2 \theta = A^2 B^2 (1 - \cos^2 \theta) = A^2 B^2 -$   
 $A^2 B^2 \cos^2 \theta$ .

**4.2**  $\boxed{U^2 V^2 \cos^2 \theta + U^2 V^2 \sin^2 \theta} = U^2 V^2$ .

**4.3 (a)**  $|\mathbf{u} \times \mathbf{v}| = 2 \cdot \sqrt{6} = \boxed{2\sqrt{6}} \approx 4.90$ . **(b)**  
 $|\mathbf{v}|^2 = 11$  and  $3^2 + 24 = 11|\mathbf{u}|^2$ :  $\boxed{|\mathbf{u}| = \sqrt{3}} \approx$   
1.73. **(c)**  $3p + q = 5$  and  $p^2 + (q - 1)^2 = 2$  give  
 $5p^2 - 12p + 7 = 0$ :  $\boxed{p = 1, q = 2}$  or

$$\boxed{p = \frac{7}{5}, q = \frac{4}{5}}.$$

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## § 5

**5.1**  $(1, 2, 1) \cdot (1, -1, -2) = -3$ , both lengths  
 $\sqrt{6}$ :  $\cos \theta = \frac{3}{6} = \frac{1}{2}$ ,  $\boxed{\theta = 60^\circ}$ .

**5.2**  $(-1, 1, -1) \cdot (5, 1, -7) = 3$ , lengths  $\sqrt{3}$  and  $5\sqrt{3}$ :  $\boxed{\cos \theta = \frac{1}{5}}$ .

**5.3**  $(3, 2, -1)$  and  $(4, \cos \alpha, \sin \alpha)$ , lengths  $\sqrt{14}$  and  $\sqrt{17}$ :  $12 + 2 \cos \alpha - \sin \alpha = \sqrt{238} \sin \alpha$ , so  $\boxed{\alpha = 0.932}$ .

**5.4**  $\mathbf{a} \times \mathbf{n} = (0, 0, 36)$  and  $(0, -36, 0) \cdot (0, 0, 36) = 0$ :  $\boxed{90^\circ}$ .

**5.5**  $\cos \widehat{BOM} = \frac{-9.721}{36}$ ,  $\widehat{BOM} = 1.844$  rad, arc  $6 \times 1.844 = \boxed{11.1}$  thousand km.

**5.6**  $\mathbf{b} \times \mathbf{m} = (-15.15, -26.23, 16.84)$ ,  $\mathbf{b} \times \mathbf{p} = (-18, -31.18, 0)$ :  $\cos \beta = 0.8740$ ,  $\boxed{029.1^\circ}$ .

## § 6

**6.1 (a)**  $\overrightarrow{PN} = (3 + 5\mu, -5 - 2\mu, 15 - \mu)$  and  $\overrightarrow{AB} = (5, -2, -1)$ :  $\boxed{10 + 30\mu}$ . **(b)**  $\mu = -\frac{1}{3}$ :  $\boxed{N\left(\frac{1}{3}, -\frac{10}{3}, \frac{7}{3}\right)}$ .

**6.2**  $\overrightarrow{OP} \cdot (1, -2, -2) = 9\lambda - 12 = 0$ ,  $\lambda = \frac{4}{3}$ :  $\boxed{P\left(\frac{4}{3}, -\frac{2}{3}, \frac{4}{3}\right)}$ .

## § 7

**7.1**  $|\overrightarrow{PA}| = 6\sqrt{2}$  and  $d = 6\sqrt{2} \sin \frac{\pi}{3} = \boxed{3\sqrt{6}} = \sqrt{54}$ .

**7.2** Parallel:  $\overrightarrow{AB} = (-1, -2, 5)$ ,  $\overrightarrow{AB} \times (1, -1, 1) = (3, 6, 3)$ ,  $d = \frac{\sqrt{54}}{\sqrt{3}} = \boxed{3\sqrt{2}} = \sqrt{18}$ .